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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAREXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Course (MDC) – **MG2MDCHIS101** – HUMAN AND ENVIRONMENT
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Part A

Short Answers - Answer any 5 questions, each question carries 2 marks

1. Late Holocene Rainforest Crisis [U] / [CO1] - Beginning in the 6th century BCE and continuing into the early centuries of the Christian Era, the central African regions experienced significant climatic fluctuations. Environmental scholars have termed this phenomenon the 'Late Holocene Rainforest Crisis.' This climatic event led to a marked decline in the density of rainforest cover extending from present-day Cameroon to the Congo Basin, resulting in the transformation of large portions of this area into savanna woodlands.

2. Define Environment [K] / [CO1] - Environment is derived from the French word *Environner*, which mean encircle or surrounding.Environment literally means Surrounding in which we are living. Environment includes all those things on which we are directly or indirectly dependent for our survival, whether it is living component like animals, plants or non-living component like soil, air water. Environmental Protection Act (1986) defined "Environment as the sum total of water, air and land, their interrelationship among themselves and with the human beings, other living beings and property."

3. What is meant by Drain of Wealth[U] / [CO2] - The Drain of Wealth theory, proposed by Dadabhai Naoroji in Poverty and Un-British Rule in India (1901), referred to the systematic transfer of resources from India to Britain under colonial rule, contributing to India's impoverishment. It was among the earliest economic critiques of British imperialism and laid the groundwork for Indian economic nationalism, influencing leaders like Gandhi and Nehru. Bipan Chandra later affirmed its significance, presenting it as a structural feature of colonialism and a key factor in India's economic underdevelopment, supported by empirical analysis rather than sentiment.

4.List the major bovine species present in the Indian subcontinent duringthe Pleistocene age[U] / [CO2] - During the Pleistocene age, several major bovine species were present in the Indian subcontinent. Bos indicus (Indian or Zebu cattle) likely emerged towards the end of this period and became an important domestic species. The Bosprimigenius (Aurochs), the wild ancestor of modern cattle, roamed the region before being domesticated. The Bubalusbubalis (Wild Water Buffalo) was native to the subcontinent and is the ancestor of the domesticated water buffalo. Additionally, Bison bison (Bison), large herbivores found in forested areas, and the Gaur (Bosgaurus), also known as the Indian bison, were present. The Gaur continues to exist today in parts of India and Southeast Asia. These bovine species, both wild and early domesticated forms, were significant components of the region's fauna during the Pleistocene.

5.Gangetic Doab[U] / [CO2] - The Gangetic Doab refers to the fertile region between the Ganges and Yamuna rivers in northern India. This area, rich in alluvial soil, has been a cradle for early Indian civilizations, providing ideal conditions for agriculture. During the early historical period, the Doab supported extensive cultivation, particularly of rice, wheat, and barley, which contributed to the growth of urban settlements and complex societies. The region's environment was characterized by dense forests, wetlands, and riverine ecosystems, supporting diverse wildlife. Over time, the natural environment faced human intervention through deforestation, agricultural expansion, and urbanization. The Gangetic Doab became a key centre for early Indian kingdoms, contributing significantly to the subcontinent's cultural and economic development.

6.Kerala Sastra Sahitya Parishad [K] / [CO4] - Kerala Sastra Sahitya Parishad (KSSP), founded in 1962, is a leading people's science movement in Kerala. Initially focused on promoting science in Malayalam, it grew into a major force advocating scientific temper, environmental protection, and social justice. Known for the slogan "Science for Social Revolution", KSSP played a key role in the People's Campaign for Decentralized Planning and remains active in public education and grassroots activism.

7. Commons [U] / [CO4] - In the context of "Capital and Environment", the term "Commons" refers to resources that are shared by a community and not privately owned, such as forests, water bodies, pastures, and air. These resources are collectively managed by local communities, often through traditional practices, and are essential for sustaining livelihoods. However, the rise of capitalism has led to the privatization and commodification of many commons, leading to overexploitation and environmental degradation, a phenomenon known as the "tragedy of the commons". This occurs when individual interests override collective well-being, depleting shared resources. The environmental impact of privatizing the commons highlights the tension between economic development and sustainable resource management, emphasizing the need for inclusive governance systems that balance economic growth with environmental conservation.

8. Green Revolution [U] / [CO5] - The Green Revolution in India refers to a period of rapid agricultural transformation that began in the late 1960s, aimed at achieving food self-sufficiency. It involved the introduction of high-yielding varieties (HYVs) of seeds, especially for wheat and rice, along with the increased use of chemical fertilizers, pesticides, and irrigation techniques. It was supported by government policies, subsidies, and international assistance, particularly from institutions like the Ford Foundation and Rockefeller Foundation.

Part B

Short Essay - Answer any 4 questions - Each question carries 5 marks

9. Analyse the features of agrarian mode of resource use.[U] / [CO1] - The agrarian mode of resource use, which began about 10,000 years ago, marked the shift to settled farming and animal domestication. Powered by human and animal labour, early agriculture relied on simple tools and irrigation, later evolving with fossil fuels. Families cultivated land while forests and pastures remained communal. Surplus production enabled trade and urban growth, while villages organized work by age, gender, and skill. Guided by tradition, these societies viewed nature as powerful yet partly controllable. Though moderate, their activities caused deforestation, soil depletion, and species loss.

10. Why did man view nature as a source of sustenance and as a spiritualentity in pre-modern societies? [U] / [CO1] - In pre-modern societies, people viewed nature both as a source of sustenance and as a spiritual entity because their survival was directly and visibly tied to the natural world. Nature provided food, water, shelter, and materials for tools and fuel, making it essential for daily life. At the same time, its forces—like storms, droughts, floods, and seasons—were powerful, unpredictable, and beyond human control. This dual dependence and vulnerability led people to see nature not just as a resource but as something sacred or inhabited by divine forces. Rituals, beliefs, and customs emerged to honour, appease, or harmonize with these forces, blending practical respect with spiritual reverence.

11. Examine the development of agriculture and animal domesticationduring the Indus Civilization [U] / [CO2] - The development of agriculture and animal domestication in the Indus Valley Civilization began around 8000 BCE, as per Irfan Habib. Climatic changes after the last Ice Age, with increased rainfall and forest expansion, supported early settlements. The fertile floodplains of the Indus River enabled the cultivation of wheat and barley, enhanced by irrigation and plough use, allowing both Rabi and Kharif crops. Alongside farming, cattle, sheep, and goats were domesticated, providing labour, milk, meat, and hides. This integration of agriculture and animal husbandry ensured the sustainability and growth of the Indus Civilization.

12. Observe the role of Indian timber in the British military and war fronts [U] / [CO2] - Indian timber was crucial for the British military during the colonial period. It was used in shipbuilding, particularly for naval warships, and in constructing railway infrastructure that supported military logistics. Timber also played a role in building military depots and storage facilities, and during both World Wars, it was used as fuel for steam engines. Overall, Indian timber was essential for maintaining the British Empire’s military operations and infrastructure.

13. Examine the early agricultural developments in the Indian subcontinentduring early historical period [U] / [CO2] - Early agricultural developments in the Indian subcontinent began with the Indus Valley Civilization (2600-1900 BCE), which featured advanced techniques like irrigation and crop rotation, supporting crops such as wheat, barley, and cotton. In the Vedic period (1500-500 BCE), the use of iron tools like plows boosted agricultural productivity and introduced new crops such as rice and millet. Agriculture shifted from subsistence farming to supporting urban centres, with texts from the time emphasizing land and irrigation systems. This period laid the foundation for the agrarian economy in ancient India.

14. What is Old World and New World in Columbian Exchange? [U] / [CO4] - In ‘The Columbian Exchange’ (1972), historian Alfred W. Crosby defines the Old World as Europe, Asia, and Africa, and the New World as the Americas. Following Columbus’s 1492 voyage, a vast exchange of plants, animals, and diseases began between these two worlds. The New World introduced crops like maize, potatoes, tomatoes, cacao, and tobacco to the Old World, transforming global diets and agriculture. In return, the Old World brought wheat, sugarcane, horses, pigs, cattle, and deadly diseases such as smallpox and measles to the New World. This exchange drastically altered ecosystems and societies worldwide, leading to profound population decline among Indigenous Americans due to the introduction of Old-World pathogens.

15. What is the connection between New World Slavery and Plantations? [U] / [CO4] - New World slavery and plantations were deeply interconnected. European colonists established plantations in the Americas to grow labour-intensive cash crops like sugar, tobacco, and cotton. To supply the massive labour these plantations required—especially after the collapse of Indigenous populations—millions of enslaved Africans were brought through the transatlantic slave trade. Slave labour was essential to the profitability of plantations, while plantation agriculture drove the demand for more slaves. This created a self-reinforcing system that shaped the economy and society of the New World for centuries.

Part C - Essay Questions - Answer any 2 questions - Each question carries 10 marks

16.Was the pre-modern human a great conservator of the environment?Examine.[U] / [CO1] - Pre-modern humans are often seen as closer to nature, but their role as environmental conservators is complex. Scholars like Ramachandra Guha and Richard Grove argue that many indigenous and pre-modern societies practiced sustainable resource management through techniques like agroforestry, sacred groves, and water conservation, often guided by spiritual and communal values. However, these practices were not always sustainable; population pressures and ecological changes sometimes led to degradation. Grove emphasizes how colonialism disrupted traditional systems, replacing them with exploitative models that worsened environmental harm. Thus, while pre-modern humans demonstrated ecological wisdom, they were not universally great conservators of the environment.

17.Examine the post-colonial Indian states approach towardsenvironmental conservation with special reference to the Project Tiger[An] / [CO2] - Post-colonial India's approach to environmental conservation initially prioritized economic development, leading to deforestation and wildlife decline. However, by the 1970s, rising environmental concerns prompted policy changes. The government enacted key laws like the Wildlife Protection Act (1972) and Forest Conservation Act (1980) to safeguard natural resources. A major initiative was Project Tiger (1973), launched to protect the endangered tiger by creating reserves, curbing poaching, and preserving habitats. It marked a shift towards ecosystem-based conservation but also faced criticism for displacing tribal communities. Over time, India's conservation strategy expanded to include projects for elephants, crocodiles, and biosphere reserves, reflecting a more comprehensive, though sometimes contested, approach to balancing ecological protection with development needs.

18. Examine the salient features of the hunting mode of resources use.Evaluate the aspects of technology, economy, social organization andecological impacts of the hunting mode.[U] / [CO1] - The hunting mode of resource use is characterized by small, kin-based groups subsisting on local resources within a few hundred square kilometres. Their technology is simple and specialized—stone tools and basic implements—divided largely along gender lines. Economically, these societies operate a natural economy with minimal surplus, little trade, and egalitarian access to resources, reflecting an absence of private property and hierarchy. Social organization is decentralized, with leadership based on knowledge rather than coercion, and labour divided by age, sex, and skill. Ecologically, hunter-gatherers have a minimal environmental impact, maintaining sustainable practices through spiritual beliefs that see humans as part of a broader natural community. Their low population density, mobile lifestyle, and culturally embedded conservation practices ensured long-term ecological balance.

19. Write an essay on the Sardar Sarovar Dam and the environmentalissues related with it. [U] / [CO4] - The Sardar Sarovar Dam, built on the Narmada River in Gujarat, is one of India’s largest and most controversial infrastructure projects. Aimed at providing irrigation, drinking water, and hydroelectric power to multiple states, the dam has also triggered significant environmental and social concerns. Large-scale submergence of forests, agricultural land, and wildlife habitats has led to loss of biodiversity and disruption of ecosystems. The displacement of thousands of tribal and rural communities has raised serious human rights and rehabilitation issues. Additionally, the alteration of river flow has impacted downstream fisheries and wetlands. Environmentalists argue that the cumulative ecological damage and social costs outweigh the claimed benefits, making the dam a focal point for debates on sustainable development in India.